

Chapter 16: The Universe

Textbook Exercise Solutions & Answers



1. Name these.

(a) Birthplace of stars.

Nebulae

(b) The biggest planet in the Solar System.

Jupiter

(c) The galaxy which is our neighbour.

Andromeda

(d) The brightest planet in the solar system.

Venus

(e) Planet with the largest number of satellites.

Jupiter

(f) Planet without a single satellite.

Venus and Mercury

(g) Planet with a rotation different from other planets.

Venus

(h) A celestial body that carries a tail along.

Comet

2. Fill in the blanks.

(a) The group of galaxies of which our Milky Way is a part is called

Local group

(b) Comets are made of

ice and dust particles

(c) The planet appears as if it is rolling along its orbit.

Uranus

(d) is a stormy planet.

Jupiter

(e) The pole star is the best example of a type of star.

variable

3. Say if the statements given below are right or wrong. Rewrite the statements after correcting them.

(a) Venus is the planet closest to the sun.

Wrong: Mercury is the planet closest to the sun.

(b) Mercury is called a stormy planet.

Wrong: Jupiter is called a stormy planet.

(c) Jupiter is the biggest planet.

Right

4. Answer the following.

(a) What is a special characteristic of the planet Mars?

The highest and the longest mountain in the solar system 'Olympus Mons' is located on Mars.

(b) What are the type of galaxies?

The different types of galaxies identified according to their shapes are spiral, elliptical, barred spiral, irregular.

(c) Which celestial bodies does a galaxy include?

Galaxy includes stars, their planetary system, cluster of stars, nebulae, clouds of gases, clouds of dust, dead stars, newly born stars etc.

(d) Name the different types of stars.

The different types of stars are

1. Sun-like stars e.g. stars like Sirius, Alpha Centauri
2. Red Giants Stars
3. Super Nova
4. Binary or Twin Stars
5. Variable Stars e.g. Polaris (pole star)

(e) What are the types of comets and on what basis are they classified?

1. Comets are of two types: Long-period comets and short-period comets.
2. Classification of comets is based on the time (period) taken by the comets to complete one revolution around the sun.

(f) What is the difference between meteors and meteorites?

1. Meteors are rocky pieces originating from the asteroid belt. Smaller rocky pieces get completely burnt due to friction with air after they enter the earth's atmosphere.
2. Meteors which do not burn completely and fall to the surface of the earth are called meteorites.

(g) What are the characteristics of the planet Neptune?

Neptune is the eighth planet in the solar system. A season on Neptune lasts for about 41 years. On this planet winds blow with extremely high speed.

5. Match the following.

Group 'A' Group 'B' 1. Galaxy (a) From east to west 2. Comet (b) 33 satellites 3. Sun-like star (c) Spiral
4. Saturn (d) Sirius 5. Venus (e) Halley

Group 'A'	Group 'B'
1. Galaxy	(c) Spiral
2. Comet	(e) Halley
3. Sun-like star	(d) Sirius
4. Saturn	(b) 33 satellites
5. Venus	(a) From east to west